CHAPTER 25
CrashMetrics

The aim of this Chapter...

...is to explain the analysis of risk associated with market crashes. The methodology is orthogonal to that explained in the previous chapter and is used in conjunction with these other VaR methods and not instead of.

In this Chapter...

- the methodology of CrashMetrics for measuring a portfolio's exposure to sudden, unhedgeable market movements
- Platinum hedging
- crash coefficients
- margin hedging
- counterparty risk
- the CrashMetrics Index for measuring the magnitude of crashes

25.1 INTRODUCTION

The final piece of the jigsaw for estimating risk in a portfolio is **CrashMetrics**. If Value at Risk is about normal market conditions then CrashMetrics is the opposite side of the coin, it is about ‘fire sale’ conditions and the far-from-orderly liquidation of assets in far-from-normal conditions. CrashMetrics is a dataset and methodology for estimating the exposure of a portfolio to extreme market movements or crashes. It assumes that the crash is unhedgeable and then finds the worst outcome for the value of the portfolio. The method then shows how to mitigate the effects of the crash by the purchase or sales of derivatives in an optimal fashion, so-called Platinum hedging. Derivatives have sometimes been thought of as being a dangerous component in a portfolio, in the CrashMetrics methodology they are put to a benign use.

25.2 WHY DO BANKS GO BROKE?

There are two main reasons why banks get into serious trouble. The first reason is the lack of suitable or sufficient control over the traders. Through misfortune, negligence or dishonesty, large and unmanageable positions can be entered into. The consequences are either that the trader concerned becomes a hero and the bank makes a fortune, or the bank loses a fortune, the trader makes a run for it and the bank goes under. The odds are fifty-fifty. The second causes of disaster are the extreme, unexpected and unhedgeable moves in the stock market, the crashes.

25.3 MARKET CRASHES

In typical market conditions one’s portfolio will fluctuate rapidly, but not dramatically. That is, it will rise and fall, minute by minute, day by day, but will not collapse. There are times, say once a year on average, when that fluctuation is dramatic. . . and usually in the downward direction. These are extreme market movements or market crashes. VaR can tell us nothing about these and they must be analyzed separately.

What’s special about a crash? Two things spring to mind. Obviously a crash is a sudden fall in market prices, too rapid for the liquidation of a portfolio. But a crash isn’t just a rise in volatility. It is also characterized by a special relationship between individual assets. During a crash, all assets fall together. There is no such thing as a crash where half the stocks fall and the rest stay put. Technically this means that all assets become perfectly correlated. In normal market conditions there may be some relationship between stocks, especially those in the same sector, but this connection may not be that strong. Indeed, it is the weakness of these relationships that allows diversification. A small insurance company will happily insure your car, because they can diversify across individuals. Insuring against an earthquake is a different matter. A high degree of correlation makes diversification impossible. This is where traditional VaR falls down, at exactly the time when it is needed most. Figure 25.1 shows the behavior of the correlation of several constituents of the S&P500 around the time of the 1987 stock market crash.

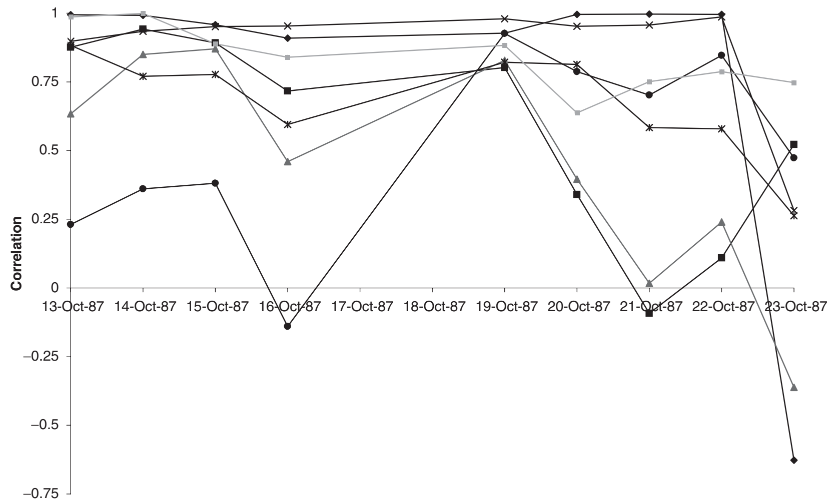


Figure 25.1 The correlation between several assets for a few days before and after the 1987 crash. When the correlation is close to one all assets move in the same direction.

All is not lost. VaR is a very recent concept, created during the 1990s and fast becoming a market standard, with known drawbacks. Many researchers in universities and in banks are turning their thoughts to analyzing and protecting against crashes. Some of these researchers are physicists who concern themselves with examining the tails of returns distributions; are crashes more likely than traditional theory predicts? The answer is a definite yes.

My personal preference though is for models that don't make any assumptions about the likelihood of a crash. One line of work is that of 'worst-case scenarios.' Given that a crash could wipe out your portfolio, wouldn't you like to know what is the worst that could realistically happen, or would you be happy knowing what you would lose on average? CrashMetrics is used to analyze worst cases, and provide advice about how to hedge or insure against a crash.

25.4 CRASHMETRICS

CrashMetrics is a methodology for evaluating portfolio performance in the event of extreme movements in financial markets. It is not part of the JP Morgan family of performance measures. We will see how in CrashMetrics the portfolio of financial instruments is valued under a worst-case scenario with few assumptions about the size of the market move or its timing. The only assumptions made are that the market move, the 'crash' is limited in size and that the number of such crashes is limited in some way. There are no assumptions about the probability distribution of the size of the crash or its timing.

This method, used for day-to-day portfolio protection, is concerned with the extreme market movements that may occur when we are not watching, or that cannot be hedged away. These are the fire sale conditions. This is the method I will explain for the rest of this chapter. There are many nice things about the method such as its simplicity and ease of generalization, and no explicit dependence on the annoying parameters volatility and correlation.

25.5 CRASHMETRICS FOR ONE STOCK

To introduce the ideas, let's consider a portfolio of options on a single underlying asset. For the moment think in terms of a stock, although we could equally well talk about currencies, commodities or interest rate products.

If the stock changes dramatically by an amount δS how much does the portfolio of options on that stock behave? There will be a relationship between the change in the portfolio value $\delta \Pi$ and δS:

$$\delta \Pi = F(\delta S).$$

The function $F(\cdot)$ will simply be the sum of all the formulæ, expressions, numerical solutions... for each of the individual contracts in the portfolio. Think of it as the sum of lots of Black–Scholes formulæ with lots of different strikes, expiries, payoffs. If there is no change in the asset price there will be no change in the portfolio, so $F(0) = 0$. (There will be a small time decay, which we'll come back to later.) Figure 25.2 shows a possible portfolio change against underlying change.

If we are lucky, and we are not too near to the expiries and strikes of the options, then we could approximate the portfolio by the Taylor series in the change in the underlying asset:

$$\delta \Pi = \Delta\ \delta S + \tfrac{1}{2}\Gamma\ \delta S^2 + \dots. \tag{25.1}$$

In practice this won't be a good enough approximation. Imagine having some knock-out options in the portfolio, we really will have to use the relevant formula or numerical method to capture the sudden drop in value of this contract at the barrier. A simple delta-gamma approximation is not going to work.

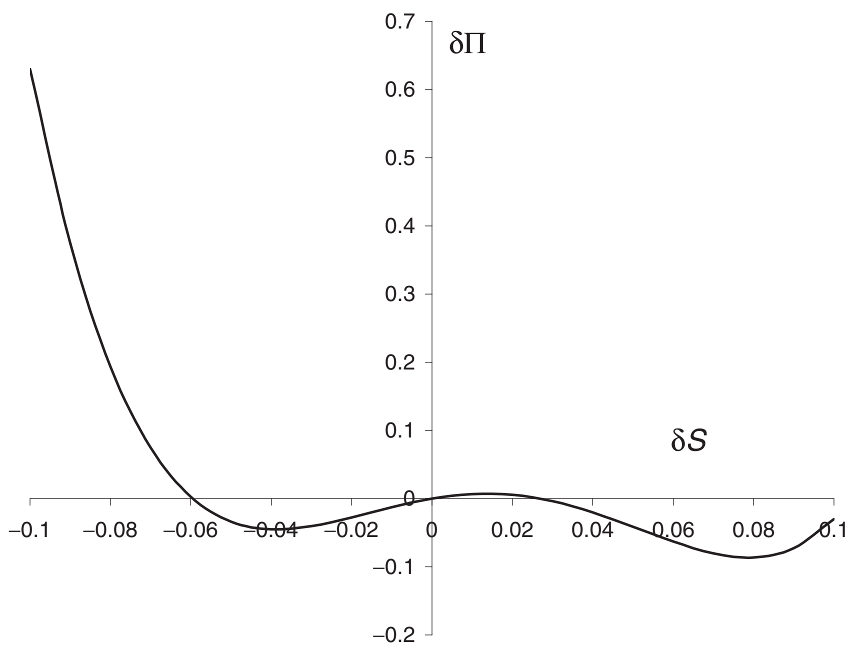


Figure 25.2 Size of portfolio change against change in the underlying.

However, as far as the math is concerned I'm going to show you both the general CrashMetrics methodology and the simple Taylor series version.

Now let's ask what is the worst that could happen to the portfolio overnight say? We want to find the minimum of $F(\delta S)$.

In practice we would only want to look at crashes/rallies of up to a certain magnitude. For this reason we may want to constrain the move in the underlying by

$$-\delta S^- < \delta S < \delta S^+.$$

If we can't use the greek approximation (Taylor series) then we're looking for

$$\min_{-\delta S^- < \delta S < \delta S^+} F(\delta S).$$

Figure 25.2 shows an example where there is one local minimum as well as a global one; it's the global one we want.

In Figure 25.3 we see a plot of the change in the portfolio against δS assuming that a Taylor approximation is valid. Note that it is zero at $\delta S = 0$. If the gamma is positive the portfolio change (25.1) has a minimum at

$$\delta S = -\frac{\Delta}{\Gamma}.$$

The portfolio change in this worst-case scenario is

$$\delta \Pi_{\text{worst}} = -\frac{\Delta^2}{2\Gamma}.$$

This is the worst case given an arbitrary move in the underlying. If the gamma is small or negative the worst case will be a fall to zero or a rise to infinity, both far too unrealistic.

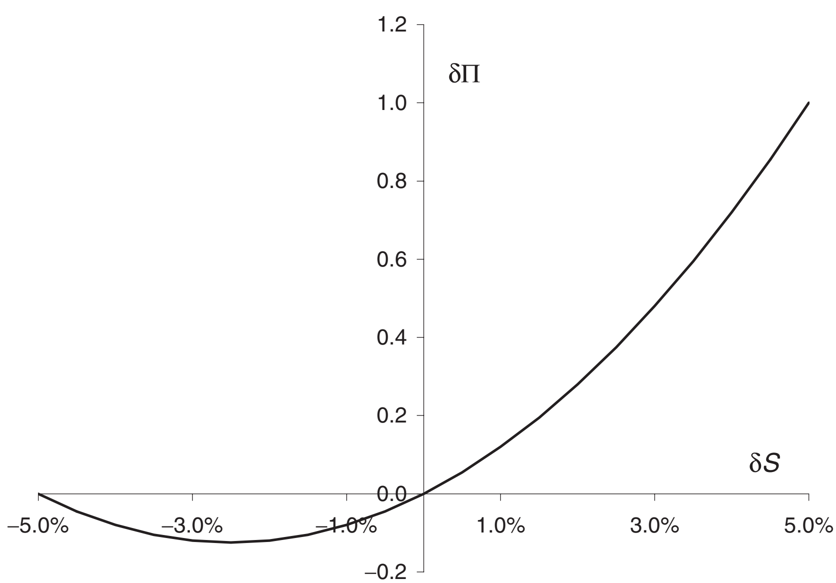


Figure 25.3 Size of portfolio change against change in the underlying, Taylor approximation.

25.6 PORTFOLIO OPTIMIZATION AND THE PLATINUM HEDGE

Having found a technique for finding out what could happen in the worst case, it is natural to ask how to make that worst case not so bad. This can be done by optimal static hedging.

To start with, I'll assume the Taylor expansion and then generalize.

Suppose that there is a contract available with which to hedge our portfolio. This contract has a bid-offer spread, a delta and a gamma. I will call the delta of the hedging contract Δ^*, meaning the sensitivity of the hedging contract to the underlying asset. The gamma is similarly Γ^*. Denote the bid-offer spread by $C > 0$, meaning that if we buy (sell) the contract and immediately sell (buy) it back we lose this amount.

Imagine that we add a number λ of the hedging contract to our original position. Our portfolio now has a first-order exposure to the crash of

$$\delta S\left(\Delta + \lambda\Delta^*\right)$$

and a second-order exposure of

$$\tfrac{1}{2}\delta S^2\left(\Gamma + \lambda\Gamma^*\right).$$

Not only does the portfolio change by these amounts for a crash of size δS but also it loses a *guaranteed* amount

$$|\lambda|\, C$$

just because we cannot close our new position without losing out on the bid-offer spread.

The total change in the portfolio with the static hedge in place is now

$$\delta\Pi = \delta S\left(\Delta + \lambda\Delta^*\right) + \tfrac{1}{2}\delta S^2\left(\Gamma + \lambda\Gamma^*\right) - |\lambda|\, C.$$

In general, the optimal choice of λ is such that the worst value of this expression for $-\delta S^- \leq \delta S \leq \delta S^+$ is as high as possible. Thus we are exchanging a guaranteed loss (due to bid-offer spread) for a reduced worst-case loss. This is simply insurance and the optimal choice gives the **Platinum hedge**, named for the plastic card that comes after green and gold.[1] For the optimal choice of the λ Figure 25.4 shows the change in the portfolio value as a function of δS. Note that it no longer goes through $(0, 0)$.

In Figure 25.5 is shown a simple spreadsheet for finding the worst-case scenario and the Platinum hedge when there is a single asset.

If we can't use the Taylor approximation, as will generally be the case, we must look for the worst case of

$$F(\delta S) + \lambda F^*(\delta S) - |\lambda| C.$$

Here $F^*(\cdot)$ is the 'formula' for the change in value of the hedging contract.

[1] Next, the Centurion hedge?

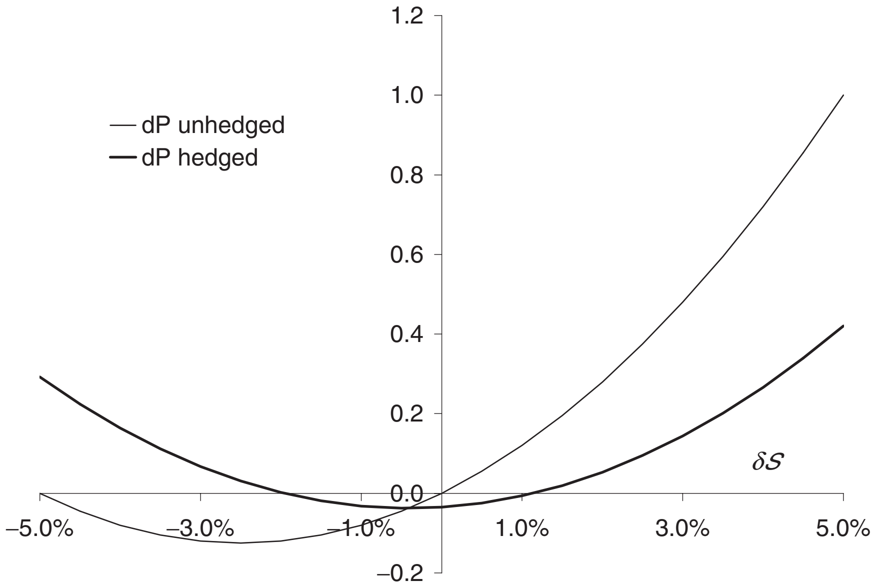


Figure 25.4 Size of portfolio change against δS after optimal hedging, Taylor approximation.

Having found the worst case, we just make this as painless as possible by optimizing over the hedge ratio λ.

Of course, there won't just be the one option with which to statically hedge, there will be many. How does this change the optimization? We'll find out soon.

25.6.1 Other 'cost' functions

In the above we have said that the 'cost' associated with the above is the bid-offer spread on the static hedge part of the portfolio. This is but one of several choices. The reason for this choice is that we may decide tomorrow to get rid of the static hedge, and hence that would genuinely be our cost.

We could also say that the cost function was the value of the bought options, in the sense that we are writing off the 'insurance.' Another choice would be to subtract a cost representing the difference between what we value the static hedge at and the value in the market. This would be suitable for a portfolio that has been constructed for its volatility arbitrage possibilities. An option bought for static hedging that is correctly priced would therefore have no associated cost.

25.7 THE MULTI-ASSET/SINGLE-INDEX MODEL

A bank's portfolio has many underlyings, not just the one. How does CrashMetrics handle them? This is done via an index or benchmark.

We can measure the performance of a portfolio of assets and options on these assets by relating the magnitude of extreme movements in any one asset to one or

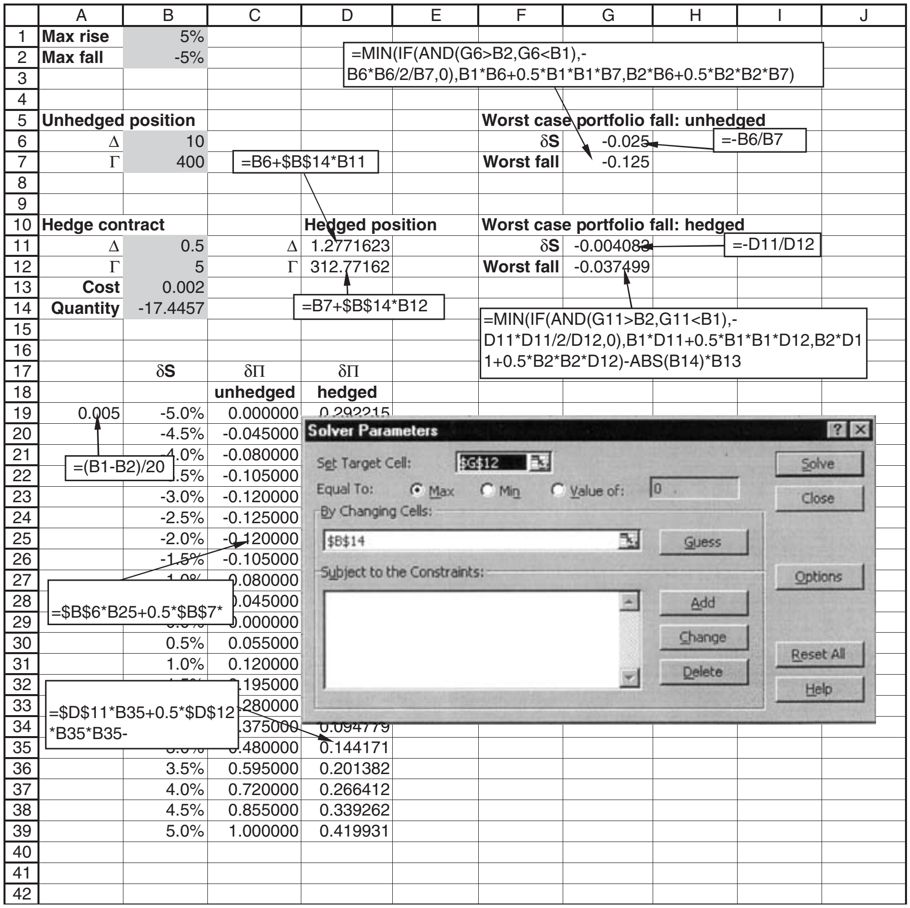


Figure 25.5 Spreadsheet for implementing basic CrashMetrics in one asset.

more **benchmarks** such as the S&P500. The relative magnitude of these movements is measured by the **crash coefficient** for each asset relative to the benchmark. If the benchmark moves by x% then the ith asset moves by $\kappa_i x$%. Estimates of the κ_i for the constituents of the S&P500, with that index as the benchmark, may be downloaded free of charge from `www.crashmetrics.com`. Note that the benchmark need not be an index containing the assets, but can be any representative quantity. Unlike the RiskMetrics and CreditMetrics datasets, the CrashMetrics dataset does not have to be updated frequently because of the rarity of extreme market movements.

The following tables give the crash coefficients for a few constituents of major indices in several countries. The crash coefficients have been estimated using the tails of the daily return distributions from the beginning of 1985 until the end of 1997, and so include the Black Monday crash of October 1987 and the rice/dragon/sake/Asian 'flu' effect starting in October 1997. For example, in Table 25.1 we see the 10 largest positive and negative daily returns in the S&P500 during that period. In this table are also shown the returns on the same days for several constituents of the index. Tables 25.2 to 25.5 show data for other markets.

Table 25.1 The 10 largest positive and negative moves in several constituents of the S&P500 against the moves in the S&P500 on the same days.

Date	S&P-500	% change	ABBOTT LABS.	ADOBE SYS.	ADVD. MICR.DEVC.	AEROQUIP-VICKERS	AETNA	AHMANSON (H.F.)
19-Oct-87	225	−20.4	−10.5	−22.2	−36.1	−36.6	−15.3	−20.8
26-Oct-87	228	−8.3	−7.3	−20.0	−14.3	−15.2	−4.5	−4.3
27-Oct-97	877	−6.9	−5.3	−6.1	−19.8	−6.7	−8.5	−6.5
08-Jan-88	243	−6.8	−3.8	−14.3	−6.8	−13.5	−7.1	−6.2
13-Oct-89	334	−6.1	−8.2	−12.5	−5.8	−9.3	−5.5	−3.7
16-Oct-87	283	−5.2	−4.6	0.0	−5.3	−6.4	−1.5	−1.3
11-Sep-86	235	−4.8	−5.2	−50.0	−4.7	−5.3	−3.3	−2.9
14-Apr-88	260	−4.4	−4.0	−12.5	−5.6	−2.6	−4.4	−4.8
30-Nov-87	230	−4.2	−6.7	−16.7	−1.4	−9.8	−2.7	−6.1
22-Oct-87	248	−3.9	−4.6	0.0	−5.9	−6.5	−1.9	−1.4
21-Oct-87	258	9.1	4.3	0.0	4.1	11.5	8.0	9.5
20-Oct-87	236	5.3	−0.6	−14.3	6.5	14.3	−3.7	3.3
28-Oct-97	921	5.1	5.2	4.3	19.0	−0.6	4.3	7.4
29-Oct-87	244	4.9	2.4	33.3	10.3	15.5	−1.9	3.2
17-Jan-91	327	3.7	4.3	0.0	4.6	8.0	2.8	3.0
04-Jan-88	255	3.6	0.8	0.0	7.6	−0.4	1.9	−0.7
31-May-88	262	3.4	2.7	0.0	5.3	0.9	3.9	2.5
27-Aug-90	321	3.2	5.4	8.3	4.5	2.2	1.4	3.1
02-Sep-97	927	3.1	3.8	2.6	3.3	0.1	2.5	2.3
21-Aug-91	391	2.9	3.1	4.2	3.5	0.0	−3.3	2.6

Table 25.2 The 10 largest positive and negative moves in several constituents of the FTSE100 against the moves in the FTSE100 on the same days.

Date	FTSE100	% change	ALLIED DOMECQ	ASDA FOODS	ASSD. BRIT	BAA	BANK OF SCOTLAND	BAR-CLAYS
20-Oct-87	1802	−12.2	−7.1	−9.0	−10.1	−7.1	−12.7	−13.2
19-Oct-87	2052	−10.8	−11.5	−9.6	−3.1	−3.7	−3.8	−11.4
26-Oct-87	1684	−6.2	−7.6	−2.4	−3.0	−3.1	−9.0	−7.5
22-Oct-87	1833	−5.7	−4.0	−7.7	−4.7	−2.6	−1.8	−1.0
30-Nov-87	1580	−4.3	−3.0	−4.3	−2.3	−3.8	−2.1	−6.3
05-Oct-92	2446	−4.1	−2.0	−2.9	2.3	−3.4	−6.8	−4.6
03-Nov-87	1653	−4.0	−3.0	−1.7	−2.0	−0.8	−2.0	−7.8
09-Nov-87	1565	−3.4	−3.9	−5.5	−1.0	−4.5	−1.6	−2.2
29-Dec-87	1730	−3.4	−2.3	−1.8	−2.3	−2.1	−0.6	−2.6
16-Oct-89	2163	−3.2	−3.2	−5.0	−3.4	−3.3	0.0	−2.2
21-Oct-87	1943	7.9	7.7	4.9	2.6	2.7	1.5	8.7
10-Apr-92	2572	5.6	8.2	3.3	3.5	7.1	13.1	7.6
17-Sep-92	2483	4.4	3.5	3.6	2.4	2.5	9.6	15.5
11-Nov-87	1639	4.2	2.3	1.8	4.4	2.1	1.0	3.4
30-Oct-87	1749	4.0	1.5	5.0	6.9	3.2	4.6	2.1
12-Nov-87	1702	3.9	1.8	−0.6	−1.0	4.1	0.8	2.2
05-Oct-90	2143	3.6	5.9	0.0	−1.0	3.2	9.3	9.8
18-Sep-92	2567	3.3	6.7	3.4	4.9	3.0	4.2	3.0
26-Sep-97	5226	3.2	1.8	−0.3	1.9	3.2	9.5	8.9
31-Dec-91	2493	3.0	4.0	7.8	1.7	1.3	4.5	2.7

Table 25.3 The 10 largest positive and negative moves in several constituents of the Hang Seng against the moves in the Hang Seng on the same days.

Date	Hang Seng	% change	AMOY PROPS	BANK OF E. ASIA	CHEUNG KONG.	CHINA LT.&POW.	FIRST PACIFIC	GREAT EAGLE
26-Oct-87	2241	−33.3	−37.4	−37.0	−29.2	−32.2	−28.3	−57.3
05-Jun-89	2093	−21.7	−36.4	−22.9	−27.0	−16.6	−28.6	−41.3
28-Oct-97	9059	−13.7	−4.8	−12.7	−9.8	−10.8	−14.7	−19.3
19-Oct-87	3362	−11.1	−18.9	0.0	−9.6	−10.5	−10.7	−20.2
22-May-89	2806	−10.8	−18.6	−8.2	−10.7	−6.2	−14.9	−15.4
23-Oct-97	10426	−10.4	−14.0	−8.9	−13.4	−7.4	−26.9	−6.3
25-May-89	2752	−8.5	−16.6	−7.6	−10.8	−5.0	−9.5	−14.9
19-Aug-91	3722	−8.4	−11.6	−4.7	−6.8	−6.8	−10.2	−9.4
03-Dec-92	4978	−8.0	−2.4	−13.4	−4.8	−9.9	−7.5	−11.3
06-Aug-90	3107	−7.4	−10.3	−6.1	−6.1	−8.2	−6.9	−6.0
29-Oct-97	10765	18.8	9.2	4.9	17.8	20.2	17.3	18.0
23-May-89	3067	9.3	11.4	7.5	9.9	5.2	11.4	14.3
06-Nov-87	2113	7.8	8.5	1.2	12.2	5.4	2.2	9.7
12-Jun-89	2440	7.6	11.7	8.7	10.5	7.0	7.0	16.9
03-Sep-97	14713	7.1	6.0	6.2	5.3	11.9	11.2	5.0
24-Oct-97	11144	6.9	4.1	2.0	9.5	5.9	12.6	7.1
27-Oct-87	2395	6.9	20.1	−2.0	3.8	9.0	−11.4	1.8
03-Nov-97	11255	5.9	5.3	7.8	7.4	−2.9	13.7	6.8
14-Jan-94	10774	5.9	6.1	3.7	6.6	3.6	3.9	4.9
19-Jun-85	1510	5.8	0.0	6.1	6.1	6.2	0.0	13.2

Table 25.4 The 10 largest positive and negative moves in several constituents of the Nikkei against the moves in the Nikkei on the same days.

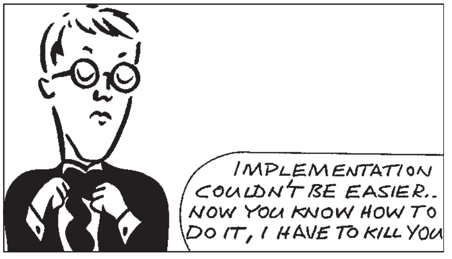
Date	Nikkei	% change	AJINO-MOTO	ALL NIPPON AIRWAYS	AOKI	ASAHI BREW.	ASAHI CHEM.	ASAHI DENKA KOGYO
20-Oct-87	21910	−14.9	−14.4	−17.9	−18.9	−18.1	−15.7	−10.8
02-Apr-90	28002	−6.6	−3.2	−6.3	−13.7	−5.0	−1.2	−5.8
19-Aug-91	21456	−6.0	−12.1	−7.9	−8.4	−1.6	−5.5	−4.7
23-Aug-90	23737	−5.8	−7.6	−12.0	−9.6	−5.4	−8.4	−7.3
23-Jan-95	17785	−5.6	−8.0	−7.2	−4.9	−1.9	−5.3	−5.2
19-Nov-97	15842	−5.3	−9.0	−3.5	−16.3	−3.4	−3.2	−5.6
24-Jan-94	18353	−4.9	−4.3	−6.0	−6.8	−1.6	−4.1	−7.6
23-Oct-87	23201	−4.9	−4.0	−6.8	−0.9	−4.1	−6.8	−3.4
26-Sep-90	22250	−4.7	−5.2	−2.4	−5.4	−1.6	−2.7	−0.6
03-Apr-95	15381	−4.7	−2.2	−4.1	−2.3	−2.0	−8.4	−6.6
02-Oct-90	22898	13.2	16.0	14.7	9.8	11.4	10.0	15.8
21-Oct-87	23947	9.3	13.5	16.3	11.6	14.7	9.3	5.2
17-Nov-97	16283	8.0	7.3	6.1	0.0	7.4	11.8	9.9
31-Jan-94	20229	7.8	8.5	3.6	13.6	4.2	4.6	9.6
10-Apr-92	17850	7.5	10.5	3.9	13.3	0.0	6.1	3.5
07-Jul-95	16213	6.3	9.8	6.3	9.2	3.0	5.0	1.2
21-Aug-92	16216	6.2	7.9	3.3	21.2	5.6	2.9	2.5
27-Aug-92	17555	6.1	5.6	−1.0	9.4	10.1	5.8	1.8
06-Jan-88	22790	5.6	5.0	6.3	6.4	2.7	6.6	2.4
15-Aug-90	28112	5.4	2.2	6.9	5.4	3.2	3.8	3.8

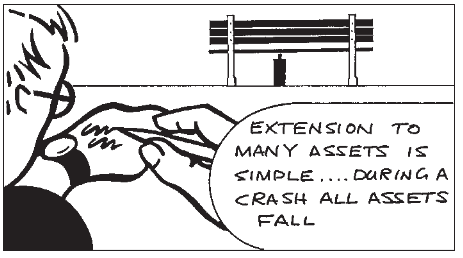
Table 25.5 The 10 largest positive and negative moves in several constituents of the Dax against the moves in the Dax on the same days.

Date	Dax	% change	ALLIANZ HLDG.	BASF	BAYER	BAYER HYPBK.	BAYERISCHE VBK.	BMW
16-Oct-89	1385	−12.8	−11.3	−10.0	−7.0	−16.1	−13.8	−13.1
19-Aug-91	1497	−9.4	−9.9	−4.8	−5.8	−11.2	−11.2	−10.0
19-Oct-87	1321	−9.4	−10.4	−9.4	−8.5	−7.0	−3.6	−8.2
28-Oct-97	3567	−8.0	−4.7	−8.3	−9.6	−8.6	−7.8	−14.8
26-Oct-87	1193	−7.7	−10.2	−4.0	−5.3	−9.0	−5.5	−6.4
28-Oct-87	1142	−6.8	−7.1	−2.6	−3.2	−8.9	−5.7	−7.5
22-Oct-87	1287	−6.7	−4.2	−4.6	−6.9	−4.6	−5.8	−7.5
10-Nov-87	945	−6.5	−8.9	−4.8	−4.1	−7.4	−6.6	−7.8
04-Jan-88	943	−5.6	−9.9	−6.9	−5.7	−2.1	−5.6	−3.3
06-Aug-90	1740	−5.4	−3.4	−4.3	−6.1	−5.5	−5.2	−6.4
17-Jan-91	1422	7.6	7.8	5.3	6.9	5.8	9.6	9.3
12-Nov-87	1061	7.4	16.6	4.7	7.6	7.6	10.8	6.4
30-Oct-87	1177	6.6	10.2	3.6	8.1	7.8	2.7	7.4
17-Oct-89	1475	6.5	5.8	3.7	2.5	5.9	9.2	5.9
01-Oct-90	1420	6.4	7.8	7.4	9.0	6.0	9.5	7.0
05-Jan-88	1004	6.4	8.6	5.1	4.7	1.7	4.6	4.4
29-Oct-97	3791	6.3	3.5	8.4	8.2	5.9	6.8	12.0
27-Aug-90	1654	6.1	3.9	8.7	5.7	5.0	2.5	6.1
21-Oct-87	1379	5.9	10.9	1.5	8.8	8.7	4.1	4.2
08-Oct-90	1465	5.3	9.1	5.0	5.2	5.1	1.8	3.7

Figure 25.6 uses the same data as used in Chapter 21 for the calculation of the beta for Disney. The fine line in this figure has slope beta. On the figure is shown the line with zero intercept that fits the largest 20 rises and falls in the S&P500, this is the bold line.

In Figure 25.7 are the returns on the Hong Kong and Shanghai Hotel group versus returns on the Hang Seng and in Figure 25.8 are the 40 extreme moves in Daimler–Benz versus returns on the Dax. It is important to note at this stage that the crash coefficient is not the same as the asset's beta with respect to the index. Not only is the number different, but preliminary results suggest that the crash coefficient is more stable than the beta. Moreover, for large moves in the index the stock and the index are far more closely correlated than under normal market conditions. In other words, when there is a crash all stocks move together.





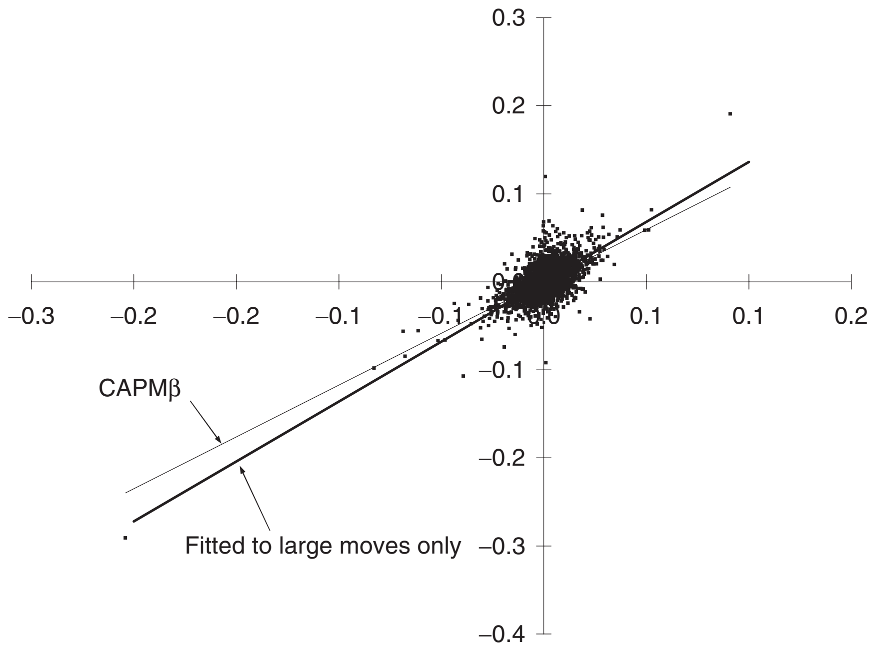


Figure 25.6 The returns on Disney versus returns on the S&P500. Also shown are the line with slope beta, fitted to all points, and the line with slope κ fitted to the 40 extreme moves and having zero intercept.

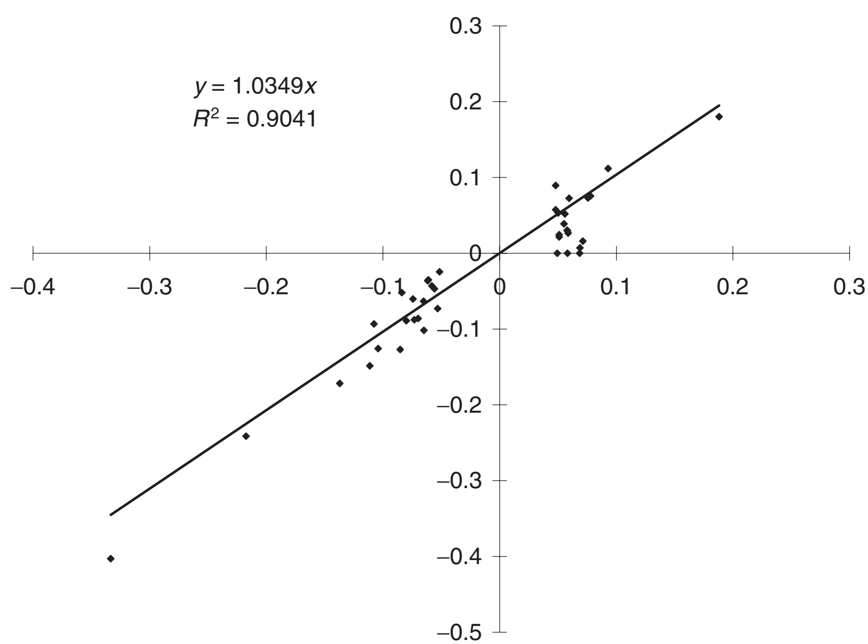


Figure 25.7 The returns on the Hong Kong and Shanghai Hotel group versus returns on the Hang Seng. Also shown is the line with slope κ fitted to the 40 extreme moves and having zero intercept.

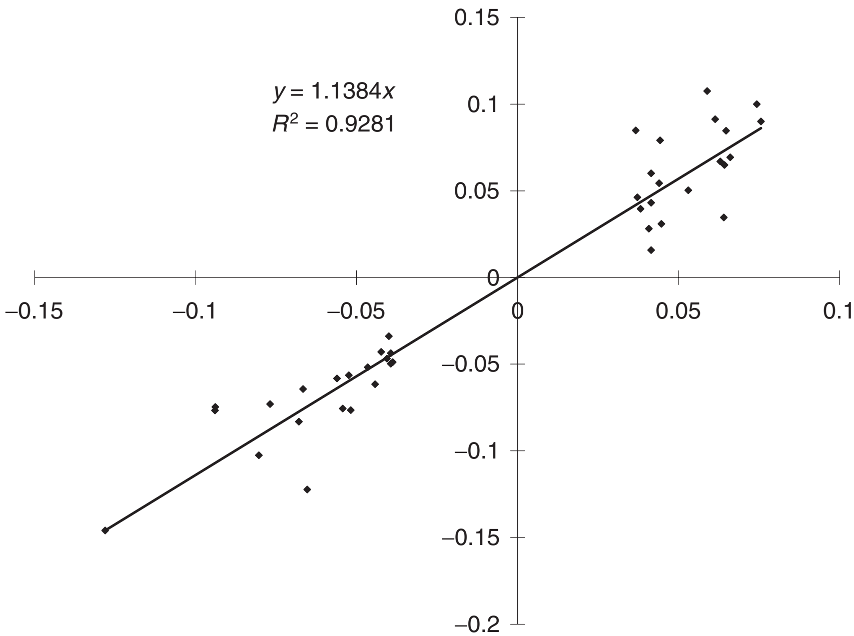


Figure 25.8 The returns on Daimler–Benz versus returns on the Dax. Also shown is the line with slope κ fitted to the 40 extreme moves and having zero intercept.

Figure 25.9 shows the different regimes for the movement of the market as a whole and individual names. There are three scenarios:

1. Typical day-to-day non-event, small movements in stock and market, perhaps some weak correlation.
2. Stock-specific event, big rise or fall in individual name, market as a whole is oblivious.
3. Market crash or dramatic rally, market and stock both move dramatically.

Because of the shape of this figure, I call it the **rings of Saturn** effect.

25.7.1 Assuming Taylor series for the moment

Let's use these ideas, first assuming a Taylor expansion for the portfolio change.

In the single-index, multi-asset model we can write the change in the value of the portfolio as

$$\delta\Pi = \sum_{i=1}^{N} \Delta_i\, \delta S_i + \tfrac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \Gamma_{ij}\, \delta S_i\, \delta S_j \tag{25.2}$$

with the obvious notation. (In particular, observe the cross gammas.) We assume that the percentage change in each asset can be related to the percentage change in the

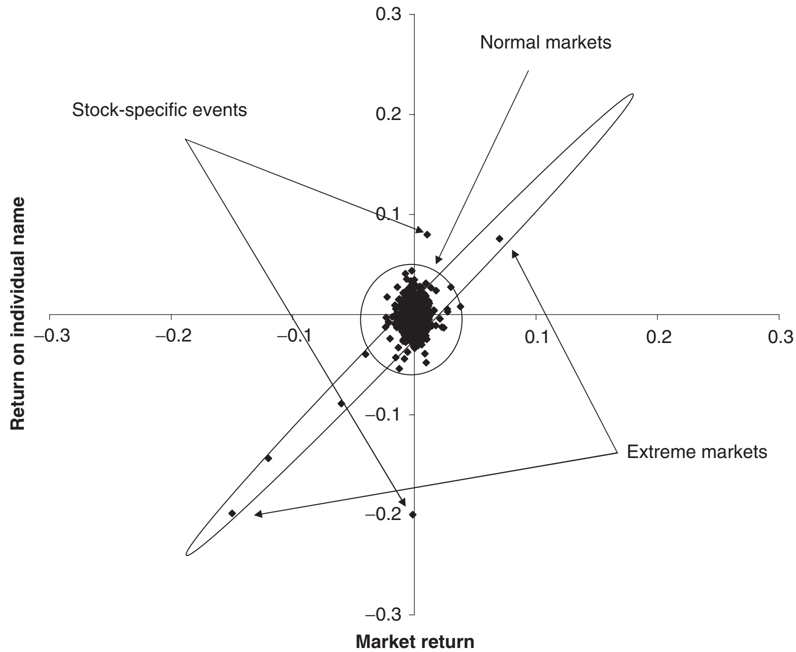


Figure 25.9 Schematic diagram showing the various returns regimes.

benchmark, x, when there is an extreme move:

$$\delta S_i = \kappa_i x S_i.$$

This simplifies (25.2) to

$$\begin{aligned}\delta\Pi &= x\sum_{i=1}^{N}\Delta_i\kappa_i S_i + \tfrac{1}{2}x^2\sum_{i=1}^{N}\sum_{j=1}^{N}\Gamma_{ij}\kappa_i S_i\kappa_j S_j\\ &= xD + \tfrac{1}{2}x^2 G.\end{aligned}$$

Observe how this contains a first- and a second-order exposure to the crash. The first-order coefficient D is the **crash delta** and the second-order coefficient G is the **crash gamma**.

Now we constrain the change in the benchmark by

$$-x^- \le x \le x^+.$$

The worst-case portfolio change occurs at one of the end points of this range or at the internal point

$$x = -\frac{D}{G}.$$

In this last case the extreme portfolio change is

$$\delta\Pi_{\text{worst}} = -\frac{D^2}{2G}.$$

We can also calculate the crash delta and gamma at this worst point.

All of the ideas contained in the single-asset model described above carry over to the multi-asset model, we just use x instead of δS to determine the worst that can happen to our portfolio.

If we can't use the delta-gamma Taylor series expansion then we must look for the worst case of an expression such as

$$\delta\Pi = F(\delta S_1, \ldots, \delta S_N) = F(\kappa_1 x S_1, \ldots, \kappa_N x S_N).$$

This is not hard, or even time consuming as long as we have formulæ for the options in our portfolio.

Suppose you want to look at the exposure of your portfolio of N underlyings in various scenarios then generally you'd want to plot an $N+1$-dimensional graph, value versus the N underlyings. However, if you want to plot the value of the portfolio in the event of a crash, using the above analysis this $N+1$-dimensional graph collapses to a more manageable two dimensions, value versus the index.

The single-index, multi-asset model

25.8 PORTFOLIO OPTIMIZATION AND THE PLATINUM HEDGE IN THE MULTI-ASSET MODEL

Suppose that there are M contracts available with which to hedge our portfolio. Let us call the deltas of the kth hedging contract Δ_i^k, meaning the sensitivity of the contract to the ith asset, $k = 1, \ldots, M$. The gammas are similarly Γ_{ij}^k. Denote the bid-offer spread by $C_k > 0$, meaning that if we buy (sell) the contract and immediately sell (buy) it back we lose this amount.[2]

Imagine that we add a number λ_k of each of the available hedging contracts to our original position. Our portfolio now has a first-order exposure to the crash of

$$x\left(D + \sum_{k=1}^{M} \lambda_k \sum_{i=1}^{N} \Delta_i^k \kappa_i S_i\right)$$

and a second-order exposure of

$$\tfrac{1}{2}x^2\left(G + \sum_{k=1}^{M} \lambda_k \sum_{i=1}^{N}\sum_{j=1}^{N} \Gamma_{ij}^k \kappa_i S_i \kappa_j S_j\right).$$

Not only does the portfolio change by these amounts for a crash of size x but also it loses a guaranteed amount:

$$\sum_{k=1}^{M} |\lambda_k|\, C_k$$

just because we cannot close our new positions without losing out on the bid-offer spread.[3]

---

[2] Of course, as explained above, other cost functions could be used.

[3] We can get an idea of which options are good value for Platinum hedging purposes by looking at the ratio $\kappa/\sigma_{\text{imp}}$. The cost of an at-the-money option is roughly proportional to its implied volatility while its effectiveness in a crash depends on its crash coefficient.

The total change in the portfolio with the static hedge in place is now

$$\delta\Pi = x\left(D + \sum_{k=1}^{M}\lambda_k \sum_{i=1}^{N}\Delta_i^k \kappa_i S_i\right) + \tfrac{1}{2}x^2\left(G + \sum_{k=1}^{M}\lambda_k \sum_{i=1}^{N}\sum_{j=1}^{N}\Gamma_{ij}^k \kappa_i S_i \kappa_j S_j\right) - \sum_{k=1}^{M}|\lambda_k|\,C_k.$$

And if we can't use the Taylor series expansion? We must examine

$$\delta\Pi = F(\kappa_1 x S_1, \ldots, \kappa_N x S_N) + \sum_{k=1}^{M}\lambda_k F_k(\kappa_1 x S_1, \ldots, \kappa_N x S_N) - \sum_{k=1}^{M}|\lambda_k|\,C_k.$$

Here F is the original portfolio and the F_ks are the available hedging contracts.

From now on I'll stick to the delta-gamma approximation and leave it to you to do the more robust and realistic whole-formulæ approach.

Platinum hedging

25.8.1 The marginal effect of an asset

We can separate the contribution to the portfolio movement in the worst case into components due to each of the underlyings:

$$\delta\Pi_i = x^*\Delta_i\,\delta S_i + \tfrac{1}{2}{x^*}^2\sum_{j=1}^{N}\Gamma_{ij}\,\delta S_i\,\delta S_j,$$

where x^* is the value of x in the worst case. This has, rather arbitrarily, divided up the parts with exposure to two assets (when the cross gamma is non-zero) equally between those assets. The ratio

$$\frac{\delta\Pi_i}{\delta\Pi_{\text{worst}}}$$

measures the contribution to the crash from the ith asset.

25.9 THE MULTI-INDEX MODEL

In the same way that the CAPM model can accommodate multiple indices, so we can have a multiple index CrashMetrics model. I will skip most of the details, the implementation is simple.

We fit the extreme returns in each asset to the extreme returns in the indices according to

$$\delta S_i = \sum_{j=1}^{n}\kappa_i^j x_j,$$

where the n indices are denoted by the j sub/superscript.

The change in value of our portfolio of stocks and options is now quadratic in all of the x_js. At this point we must decide over what range of index returns do we look for the worst case. Consider just the two index case, because it is easy to draw the pictures.

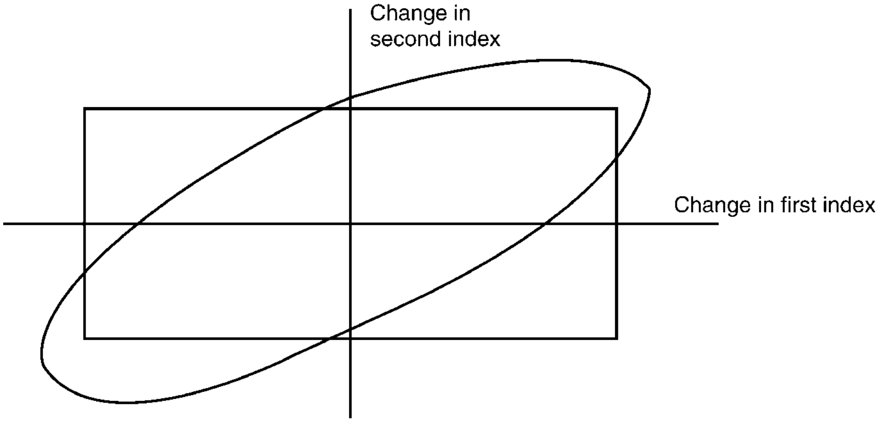


Figure 25.10 Regions of interest in the two-index model.

One possibility is to allow x_1 and x_2 to be independent, to take any values in a given range. This would correspond to looking for the minimum of the quadratic function over the rectangle in Figure 25.10. Note that there is no correlation in this between the two indices, fortunately this difficult-to-measure parameter is irrelevant. Alternatively if you believe that there is some relationship between the size of the crash in one index and the size of the crash in the other you may want to narrow down the area that you explore for the worst case. An example is given in the figure.

25.10 INCORPORATING TIME VALUE

Generally we are interested in the behavior over a longer period than overnight. Can we examine the worst case over a finite time horizon? We can expand the portfolio change in a Taylor series in both δS and δt, the time variable, to get:

$$\delta\Pi - r\Pi\,\delta t = (\Theta - r\Pi)\,\delta t + \Delta\,\delta S + \tfrac{1}{2}\Gamma\,\delta S^2. \tag{25.3}$$

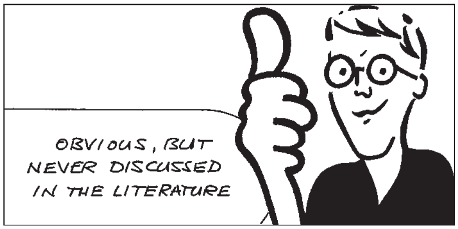
Observe that we examine the portfolio change in excess of the return at the risk-free rate. We must now determine the lowest value taken by this for

$$0 < \delta t < \tau \text{ and } -\delta S^- < \delta S < \delta S^+,$$

where τ is the horizon of interest. Since the time and asset changes decouple, the problem for the worst-case asset move is exactly the same as the above, overnight, problem. The worst-case time decay up to the horizon will be

$$\min\left((\Theta - r\Pi)\,\tau, 0\right).$$

The idea of Platinum hedging carries over after a simple modification. The modification we need is to the theta. We must incorporate the Θ^* for each of the hedging contracts, suitably multiplied by the number of contracts.



25.11 MARGIN CALLS AND MARGIN HEDGING

Stock market crashes are more common than one imagines, if one defines a crash as any unhedgeable move in prices. Although we have focused on the change in value of our portfolio during a crash this is not what usually causes trouble. One of the reasons for this is that in the long run stock markets rise significantly faster than the rate of interest, and banks are usually net long the market. What causes banks, and other institutions, to suffer is not the paper value of their assets but the requirement to suddenly come up with a large amount of cash to cover an unexpected margin call. Banks can weather extreme markets provided they do not have to come up with large amounts of cash for margin calls. For this reason it makes sense to be 'margin hedged.' Margin hedging is the reduction of future margin calls by buying/selling contracts so that the net margin requirement is insensitive to movements in underlyings. In the worst-case crash scenario discussed here, this means optimally choosing hedging contracts so that the worst-case margin requirement is optimized. Typically, over-the-counter (OTC) options will not play a role in the optimal margin hedge since they do not usually have margin call requirements.

Recent examples where margin has caused significant damage are Metallgesellschaft and Long Term Capital Management.[4]

I now show how the basic CrashMetrics methodology can be easily modified to estimate and ameliorate worst-case margin calls.

25.11.1 What is margin?

Writing options is very risky. The downside of buying an option is just the initial premium, the upside may be unlimited. The upside of writing an option is limited, but the downside could be huge. For this reason, to cover the risk of default in the event of an unfavorable outcome, the clearing houses that register and settle options insist on the deposit of a margin by the writers of options. Clearing houses act as counterparty to each transaction.

Margin comes in two forms, the initial margin and the maintenance margin. The initial margin is the amount deposited at the initiation of the contract. The total amount held as margin must stay above a prescribed maintenance margin. If it ever falls below this level then more money (or equivalent in bonds, stocks, etc.) must be deposited. The levels of these margins vary from market to market.

25.11.2 Modeling margin

The amount of margin that must be deposited depends on the particular contract. Obviously, we are not too concerned with the initial margin since this is known in advance of the purchase/sale of the contract. It is the variation margin that will concern us since a dramatic market move could result in a sudden large margin call that may be difficult to meet.

[4] The latter suffered after a 'once in a millennium...10 sigma event.' Unfortunately it happened in only their fourth year of trading.

We will model the margin call as a percentage of the change in value of the contract. We denote that percentage by the Greek letter χ. Note that for an over-the-counter (OTC) contract there is usually no margin requirement so $\chi = 0$.

With the same notation as above, the change in the value of a single contract with a single underlying is

$$\delta\Pi = \Delta\ \delta S + \tfrac{1}{2}\Gamma\ \delta S^2.$$

Therefore the margin call would be

$$\delta M = \chi\left(\Delta\ \delta S + \tfrac{1}{2}\Gamma\ \delta S^2\right).$$

The reader can imagine the details of extending this formula to many underlyings and many contracts, and to include time decay.

The final result is simply

$$\delta M - rM\,\delta t = \left(\sum_{i=1}^{N}\overline{\Theta}_i - rM\right)\delta t + \sum_{i=1}^{N}\overline{\Delta}_i\,\delta S_i + \tfrac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\overline{\Gamma}_{ij}\,\delta S_i\,\delta S_j.$$

This assumes that interest is received by the margin. Here

$$\begin{aligned}
\overline{\Theta}_i &= \textbf{margin theta} \text{ of all options with } S_i \text{ as the underlying}\\
&= \sum \chi\,\Theta, \text{ where the sum is taken over all options}\\
\overline{\Delta}_i &= \textbf{margin delta} \text{ of all options with } S_i \text{ as the underlying}\\
&= \sum \chi\,\Delta, \text{ where the sum is taken over all options}\\
\overline{\Gamma}_{ij} &= \textbf{margin gamma} \text{ of all options with } S_i \text{ and } S_j \text{ as the underlyings}\\
&= \sum \chi\,\Gamma, \text{ where the sum is taken over all options}\\
\\
r &= \text{risk-free interest rate}\\
\delta t &= \text{time horizon}\\
\delta S_i &= \text{change in value of } i\text{th asset}
\end{aligned}$$

The notation is self-explanatory.

The conclusion is that the CrashMetrics methodology will carry over directly to the analysis of margin provided that the Greeks are suitably redefined. We have therefore introduced the new Greeks, $\overline{\Theta}$, $\overline{\Delta}$ and $\overline{\Gamma}$, margin theta, margin delta and margin gamma, respectively.

The reader who is aware of the Metallgesellschaft fiasco will recall that they were delta hedged but not margin hedged.

I've assumed a Taylor series/delta-gamma approximation that almost certainly won't be realistic during a crash. We are lucky when modeling margin that virtually every contract on which there is margin has a nice formula for its price. The complex products which require numerical solution are typically OTC contracts with no margin requirements at all. I leave it to the reader to go through the details when using formulæ rather than greek approximations.

25.11.3 The single-index model

As in the original CrashMetrics methodology we relate the change in asset value to the change in a representative index via

$$\delta S_i = \kappa_i x S_i.$$

Thus we have

$$\begin{aligned}\delta M - rM\,\delta t &= \left(\sum_{i=1}^{N} \overline{\Theta}_i - rM\right)\delta t + x\sum_{i=1}^{N} \overline{\Delta}_i \kappa_i S_i + \tfrac{1}{2}x^2 \sum_{i=1}^{N}\sum_{j=1}^{N} \overline{\Gamma}_{ij}\kappa_i S_i \kappa_j S_j \\ &= \overline{Th}\,\delta t + x\overline{D} + \tfrac{1}{2}x^2\overline{G}.\end{aligned}$$

Here $\overline{Th}$ is the portfolio margin theta *in excess of the risk-free growth*. Observe how this expression contains a first- and a second-order exposure to the crash. The first-order coefficient $\overline{D}$ is the crash margin delta and the second-order coefficient $\overline{G}$ is the crash margin gamma.

We've seen worst-case scenarios, Platinum hedging and the multi-index model applied to portfolio analysis and hedging. All of these carry over unchanged to the case of margin analysis, provided that the greeks are suitably redefined. In particular Platinum margin hedging is used to optimally reduce the size of margin calls in the event of a market crash.

25.12 COUNTERPARTY RISK

If OTC contracts do not have associated margin calls, they do have another serious kind of risk: counterparty risk. During extreme markets counterparties may go broke, having a knock-on effect on other banks. For this reason, one should divide up one's portfolio by counterparty initially and examine the worst-case scenario counterparty by counterparty. Everything that we have said above about worst-case scenarios and Platinum hedging carries over to the smaller portfolio associated with each counterparty.

25.13 SIMPLE EXTENSIONS TO CRASHMETRICS

In this section I want to briefly outline ways in which CrashMetrics has been extended to other situations and to capture other market effects. Because of the simplicity of the basic form of CrashMetrics, many additional features can be incorporated quite straightforwardly.

First of all, I haven't described how the CrashMetrics methodology can be applied to interest rate products. This is not difficult, simply use a yield (or several) as the benchmark and relate changes in the values of products to changes in the yield via durations and convexities. The reader can imagine the rest.

A particularly interesting topic is what happens to parameter values after a crash. After a crash there is usually a rise in implied volatility and an increase in bid-offer spread. The rise in volatility can be incorporated into the methodology by including vega terms, dependent also on the size of the crash. This is conceptually straightforward,

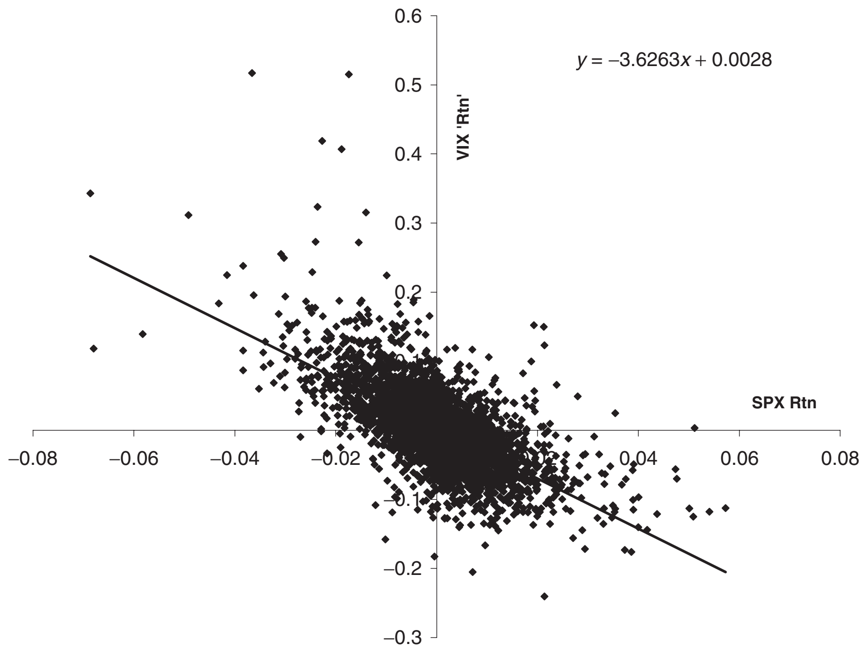


Figure 25.11 VIX return versus SPX return.

but requires analysis of option price data around the times of crashes. If you are long vanilla options during a crash, you will benefit from this rise in volatility. Similarly, a crash-dependent bid-offer spread can be incorporated but again requires historical data analysis to model the relationship between the size of a crash and the increase in the spread. See Figure 25.11.

Finally, it is common experience that shortly after a crash stocks bounce back, so that the real fall in price is not as bad as it seems. Typically 20% of the sudden loss is recovered shortly afterwards, but this is by no means a hard and fast rule. You can see this in the earlier data tables, a date on which there is a very large fall is followed by a date on which there is a large rise. To incorporate such a dynamic effect into the relatively static CrashMetrics is an interesting task.

25.14 THE CRASHMETRICS INDEX (CMI)

The results and principles of CrashMetrics have been applied to a **CrashMetrics Index** (**CMI**). This is an index that measures the magnitude of market moves and whether or not we are in a crash scenario. It's like a Richter scale for the financial world. Unlike most measures of market movements this one is *not* a volatility index in disguise, it is far more subtle than that. However, being proprietary, I can't tell you how it's defined. Sorry. I can give you some clues: it's based on a logarithmic scale; It has only one timescale (unlike volatility which needs a long timescale such as thirty days, and a short one, a day, say); it exploits the effect shown in Figure 25.1. Figure 25.12 shows a time series of the CMI applied to the S&P500.

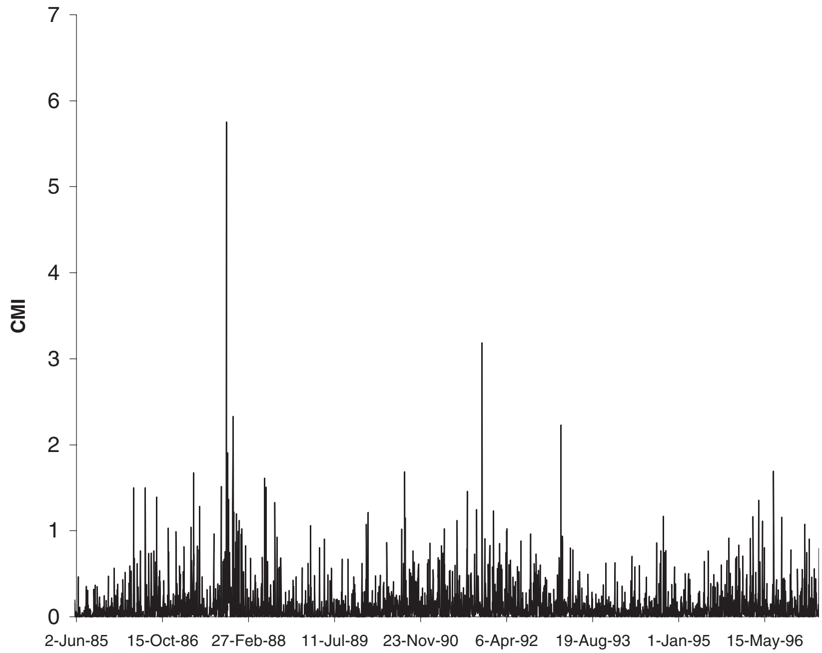


Figure 25.12 The S&P500 CMI.

25.15 SUMMARY

This chapter described a VaR methodology that is specifically designed for the analysis of and protection against market crashes. Such analysis is fundamental to the well-being of financial institutions and for that reason I have taken a non-probabilistic approach to the modeling.

FURTHER READING

- Download the CrashMetrics technical documents, datasets and demonstration software for CrashMetrics from `www.crashmetrics.com`.

EXERCISES

1. Extend the example CrashMetrics spreadsheet to incorporate many underlyings, all related via an index. How would interest rate products be incorporated?
2. Sudden, large movements in a stock are usually accompanied by an increase in implied volatilities. Incorporate a vega term into the one-asset CrashMetrics model. What is the role of actual volatility in CrashMetrics?